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Books :- S.E by Prof k.k. Aggarwal & Yogesh Singh
Roger S. Pressman.
Prof. Rankaj
Sommerville

customer ⟷ Developer
SRS (Software Requirement Specification)

→ Program + documentation + Operating Procedure = S/W
Program ⊆ S/W

→ Software has to go through phases
Software development life cycle (SDLC)

→
Build and fix Model
Waterfall Model
Spiral Model
V Model
Iterative enhancement
Evolutionary Model
RAD

① Build and fix model
2 Step model

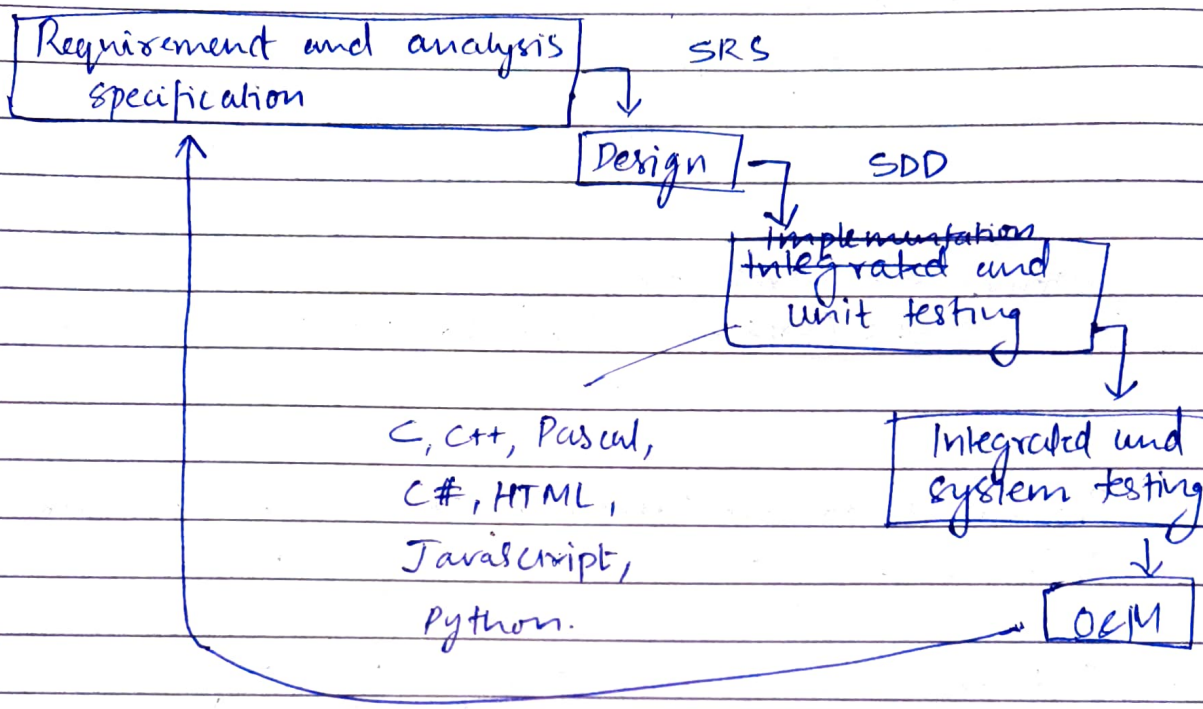


small projects

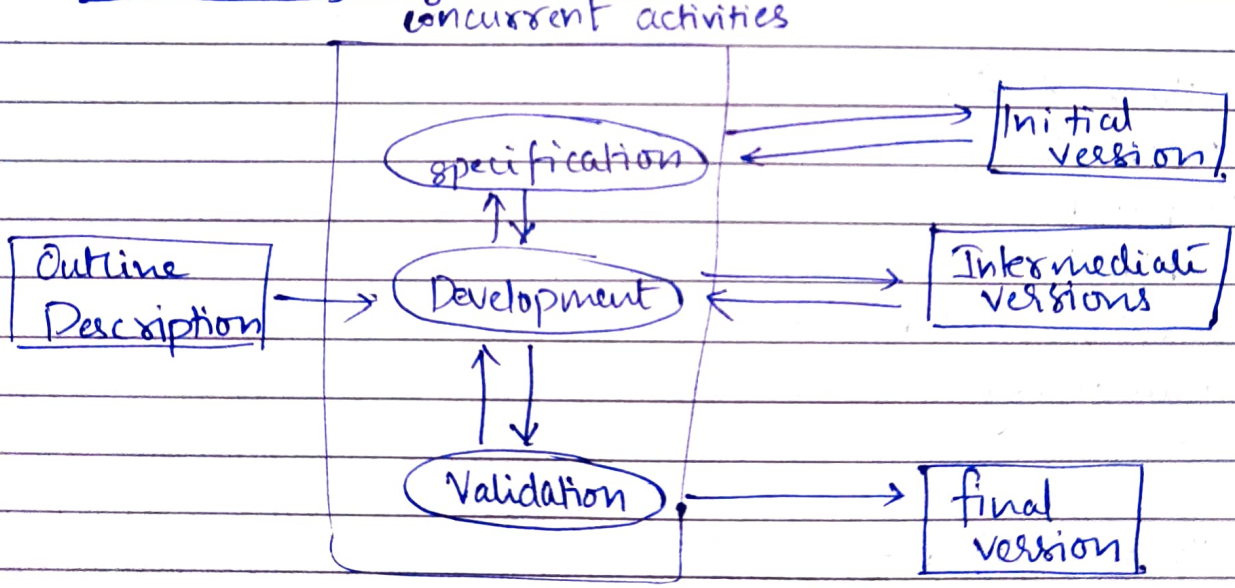
Spiral

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② Waterfall



③ Evolutionary ~~systems~~ Model



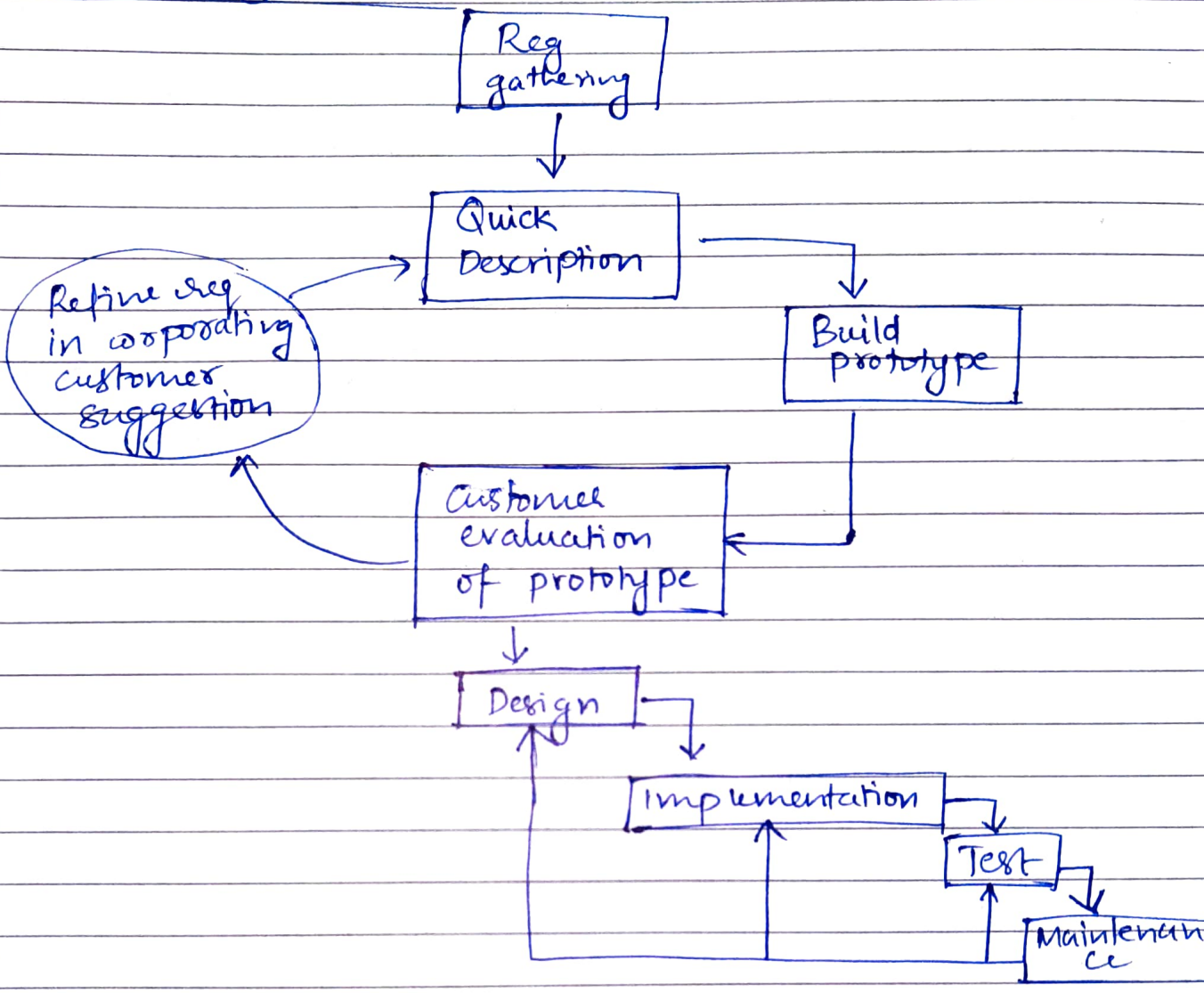
Advantages: ① Better risk analysis
② Supports changing envi
③ Initial
④ better suited for large projects
⑤ during life process software

disadvantage
→ costly to use
→ not for small projects

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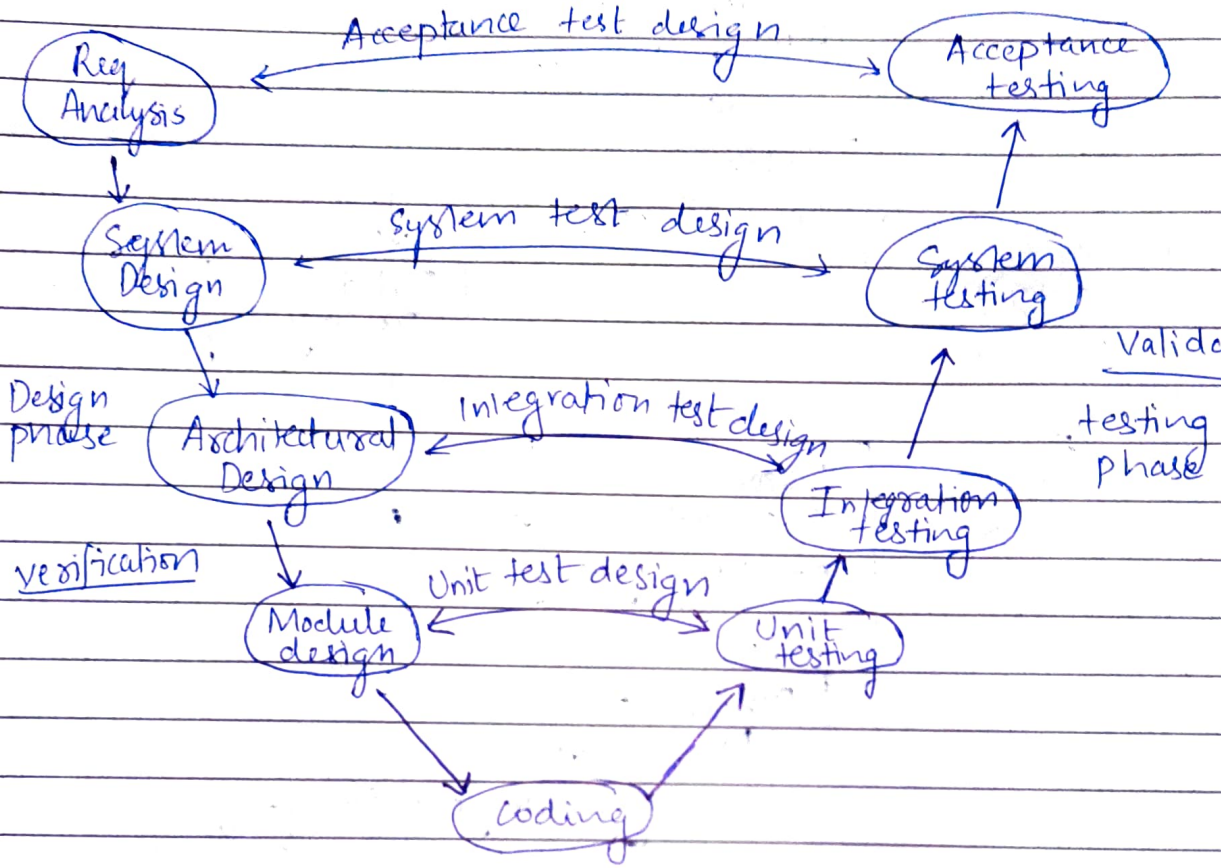
④ Prototyping Model



SPIRAL Model

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V - Model

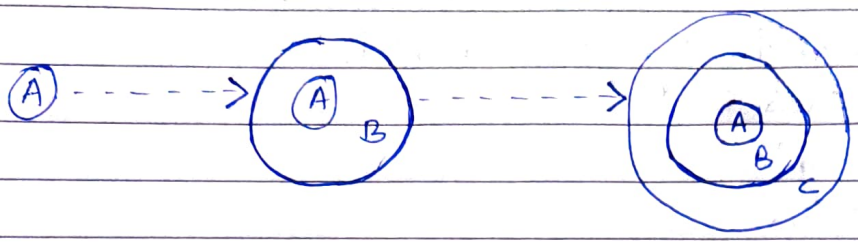


→ deliverables - end product of one phase.
→ when requirements clearly declared.
→ used in small size projects.
→ advantages : ① highly disciplined
② phases completed one at a time
③ Simple and easy understandable
④ error free, good quality software produced.
→ disadvantages : ① high risk and uncertainty
② Not good for large, object oriented projects.
③ Not good for projects that do not define the requirements clearly.

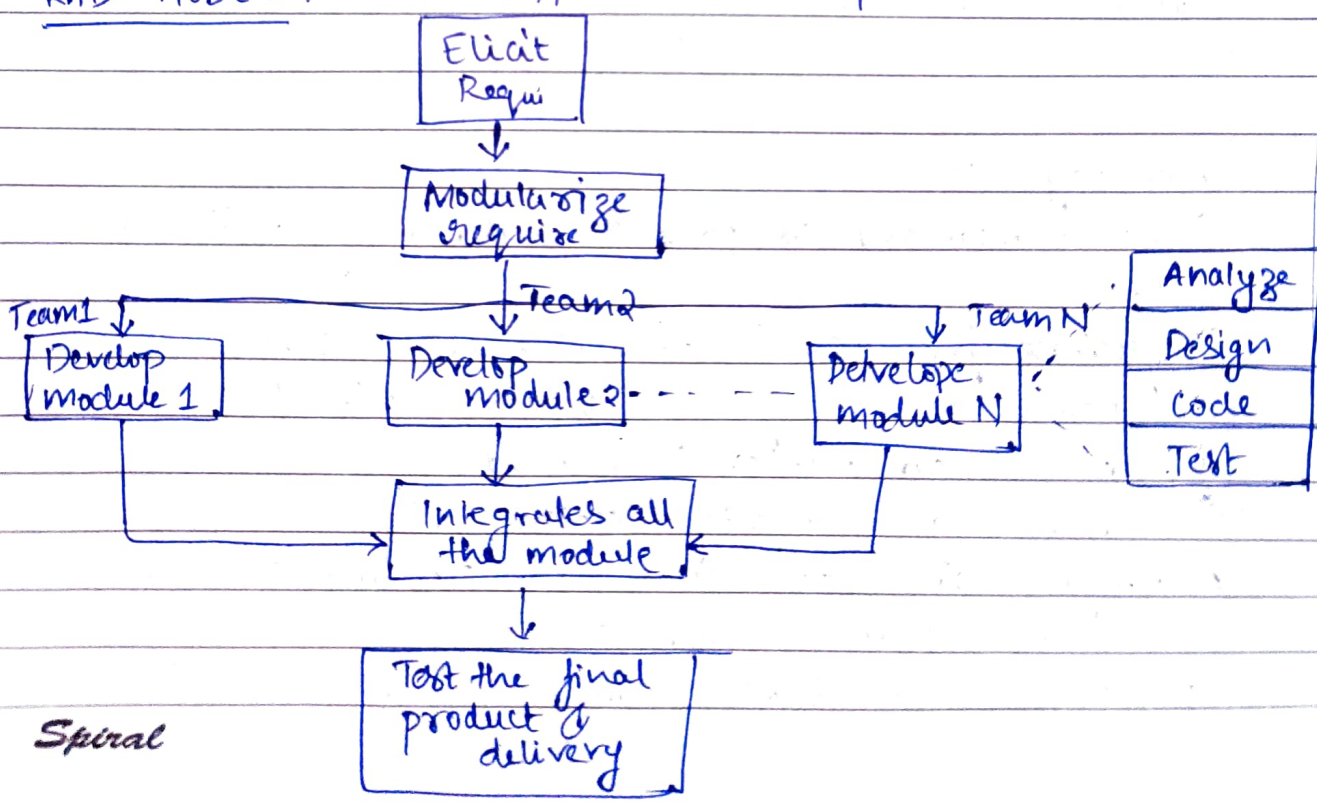
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Incremental process model :- (Successive version model)



RAD MODEL :- Rapid application development



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→ IBM in 1980 first proposed the model.
→ If a software broken into modules then it will be created by this model.
→ 4 basic phases :-
① Req^n planning
② User Description
③ Construction
④ Cut over

→ 60-90 days software can be developed by this model.
→ includes reexamination
→ prototyping
→ All the independent modules have to be tested separately.

→ When to use RAD model ?
① when customer has well known req^n and
② user is involved throughout life cycle.
③

→ Advantages :-
① use of reusable components reduces cycle time of con
② feedback from cust^m available at all stage
③ reduces cost as few developers required
④ powerful documented tools used
⑤ product good quality (optimized)
⑥ progress and development of product can be measured through various stages.
⑦ Easier to accomodate changing req^n

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→ Disadvantages:

① use of powerful and efficient tools requires highly skilled professionals.
② Absence of reusable component → leads to failure.
③ Customer involvement always required.
④ not applicable for small sized products.

→ Applications:-

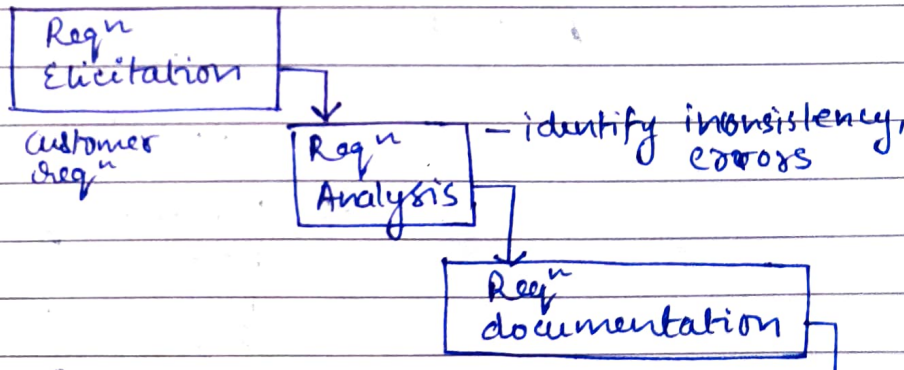
① Used for
② projects where requirements can be modularized.

06/02/23

SRS - Req^n Engg

problem statement → RE → SRS

⇒ Crucial steps in req^n engg:-



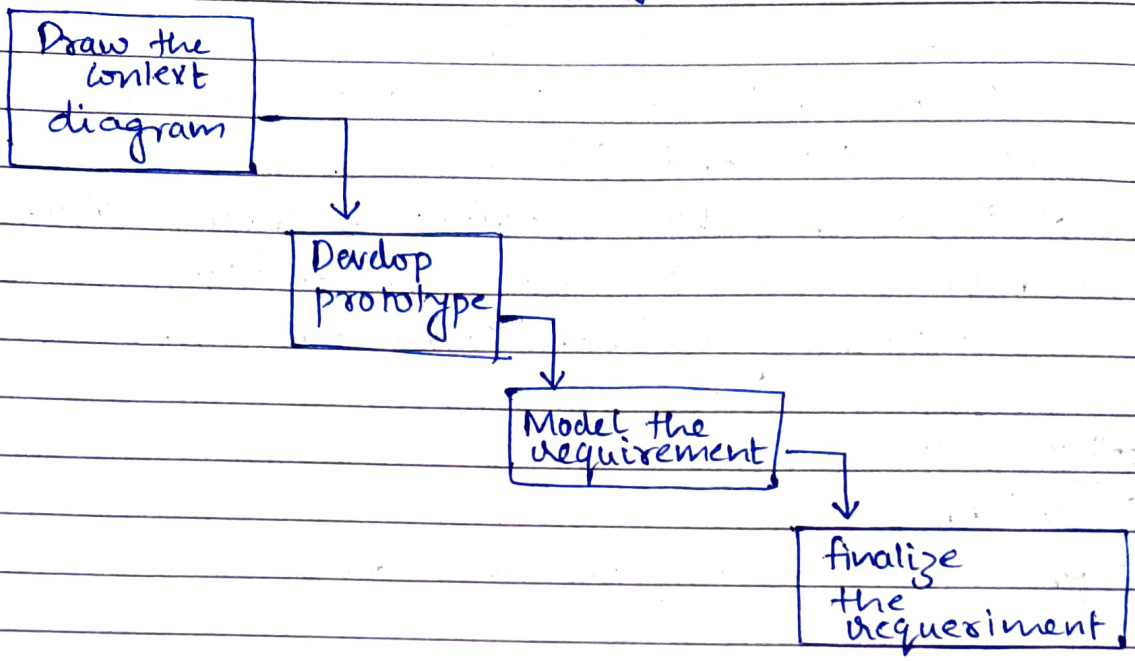
→ Requirements:-

- Functional (product features)
- Non-Functional (quality requirements)
- known requirements
- unknown requirements

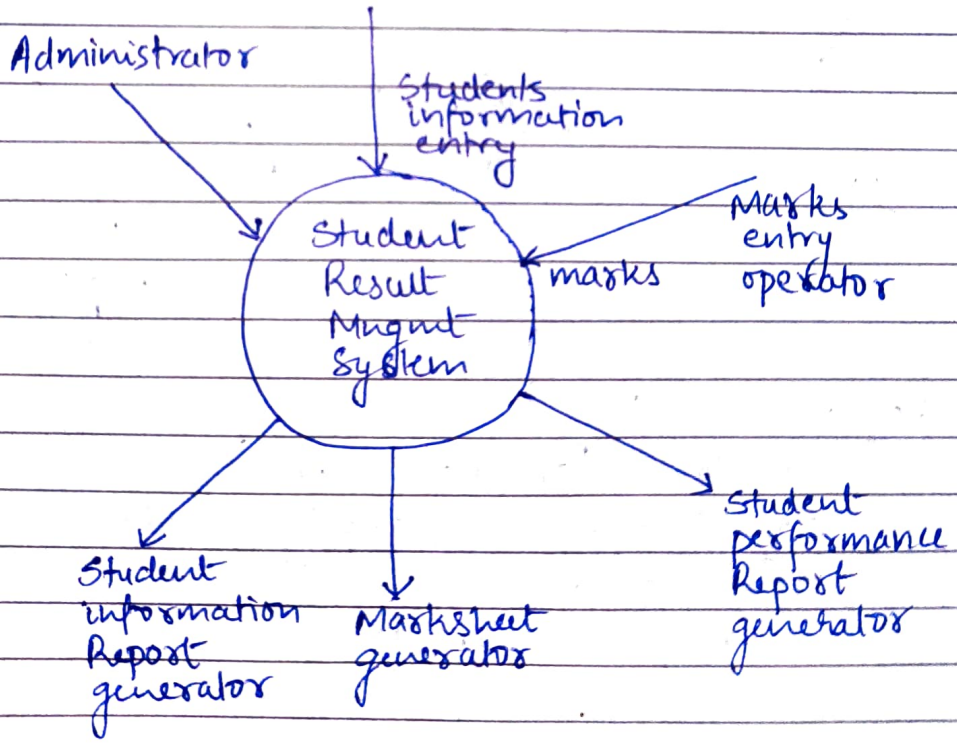
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→ Steps for requirement analysis

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eg:- develop context diagram for student mngmt system

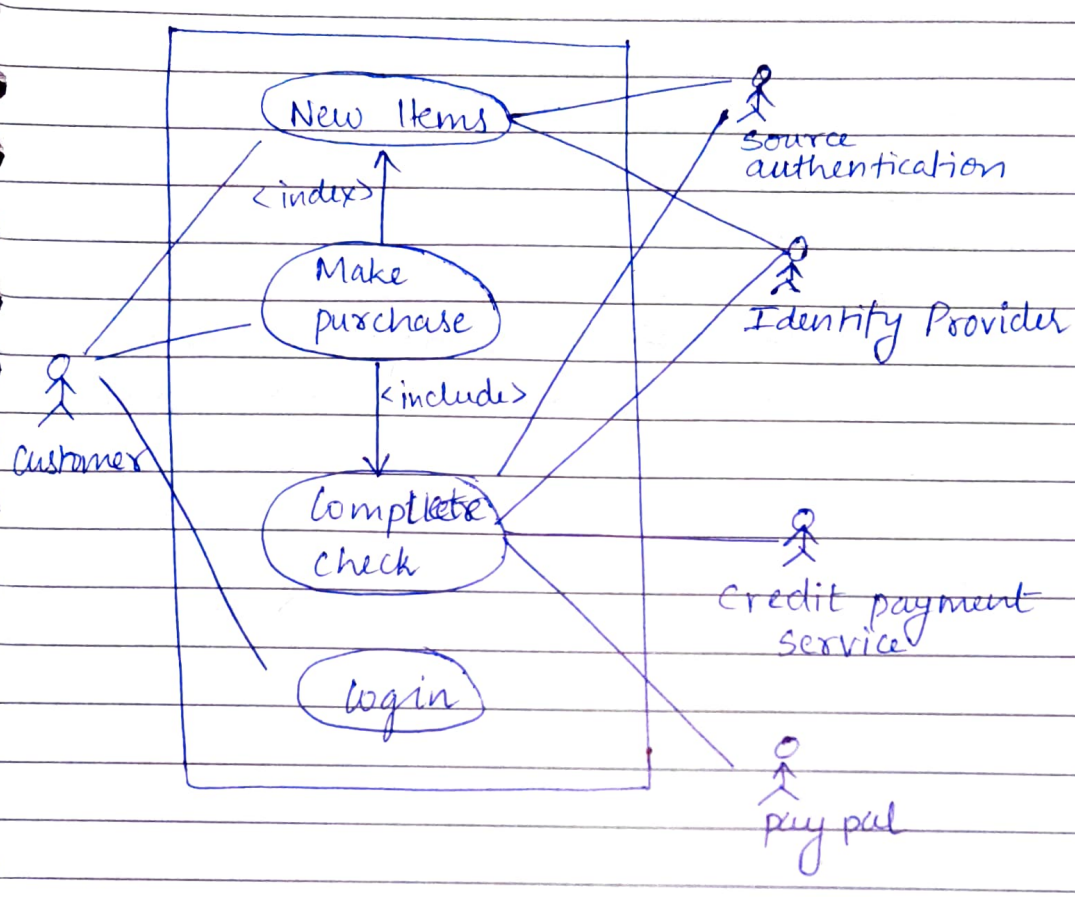


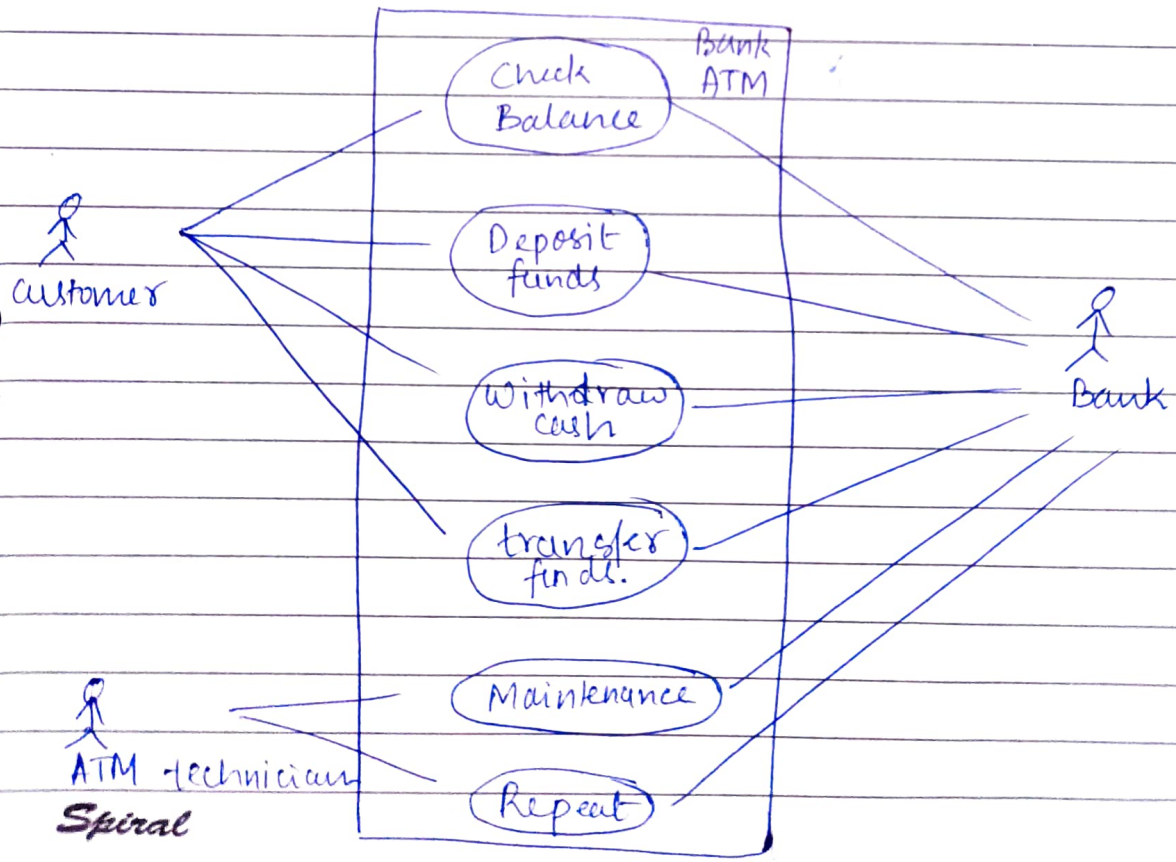
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USE-CASE DIAGRAM





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→ Use-case diagram for railway reservation system

13/02/23

Req^n Elicitation Methods

① Interview — [open ended, Structured]
② Brain storming Sessions
③ FAST (facilitated Application Specific technique)
④ QFD (Quality function Deployment) → Normal requirement, Expected req^n, Exciting Req^ng
⑤ Use case Approach

→ These 5 steps used for information gathering

Advantages:-

① helps clarify customer req
② increases chances of developing sofware sys that meets customer needs.

Disadvantages :-